

Add vectors

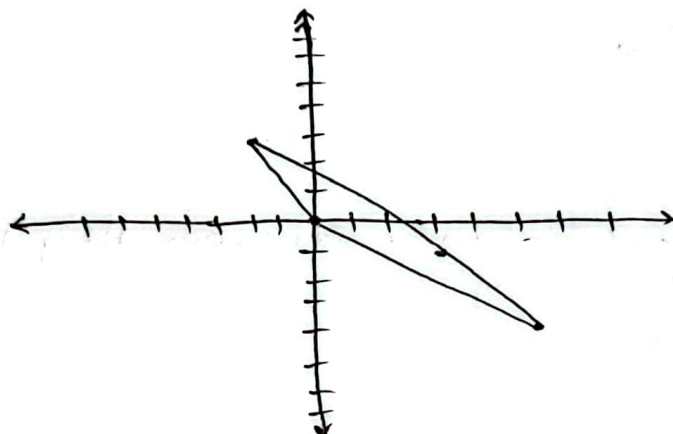
Q. These are the component forms of vectors $\vec{c}$ and $\vec{d}$:

$$\vec{c} = (5, -4)$$

$$\vec{d} = (-2, 3)$$

Answer two questions about the addition $\vec{c} + \vec{d}$.
performing the addition using the parallelogram rule, with $\vec{c}$ and $\vec{d}$ starting at the origin, creates a parallelogram on the coordinate plane

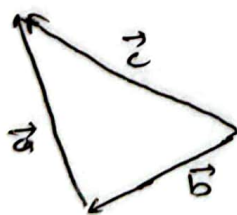
1. draw the resultant parallelogram



2. find the sum,

$$\begin{aligned}\vec{c} + \vec{d} &= (5, -4) + (-2, 3) \\ &= (3, -1)\end{aligned}$$

Q. which equation correctly describes the relationship between the vectors?



Sol.

$$\vec{a} + \vec{b} = \vec{c}$$

Q. These are the component forms of vectors $\vec{z}$ and $\vec{t}$:

$$\vec{z} = \langle -1, -4 \rangle$$

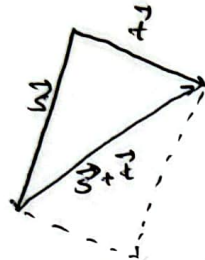
$$\vec{t} = \langle 2, -2 \rangle$$

Bodhi and Bram tried to find the sum of the vectors.

Bodhi tried adding the vectors component-wise:

$$\begin{aligned} \langle -1, -4 \rangle + \langle 2, -2 \rangle &= \langle -1+2, -4+(-2) \rangle \\ &= \langle 1, -2 \rangle \end{aligned}$$

Bram tried adding the vectors using the parallelogram rule. He drew the following diagram and concluded that the sum is $\langle 9, 2 \rangle$.



Since they didn't get the same answer, they figured at least one of them is wrong.

Who made mistake?

- Ⓐ only Bodhi
- Ⓑ only Bram
- Ⓒ Both

Ans:

Both

Q. These are the component forms of vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = \langle -4, -3 \rangle$$

$$\vec{v} = \langle -1, 5 \rangle$$

Add the vectors.

Sol:

$$\vec{u} + \vec{v} = \langle -5, 2 \rangle$$